< 2024 Creative Algorithms >

Coffee Making Flash Game

20231005 Siyun Kim

Project Summary

I created a coffee-making flash game where players aim to make the best coffee possible and then be evaluated. The game consists of six stages: **Picking**, **Roasting**, **Grinding**, **Tamping**, **Extracting**, and **Pouring**. Players need to use either a mouse or their phone to successfully complete each stage. However, since each stage progresses very quickly, players must stay focused to achieve a higher score.

Here's a breakdown of the stages:

1. Picking: Click on all the coffee beans. Failing to collect all the beans will result in a lower score.
2. Roasting: Shake the phone from bottom to top. Over- or under-roasting will negatively affect their score.
3. Grinding: Rotate the phone.
4. Tamping: Click anywhere. Skipping this step will lower their score.
5. Extracting: Simply wait for the shot to be extracted.
6. Pouring: Tilt the phone slightly. Not pouring the shot will result in a very low score.

Motivation

- **A desire to use Wekinator**

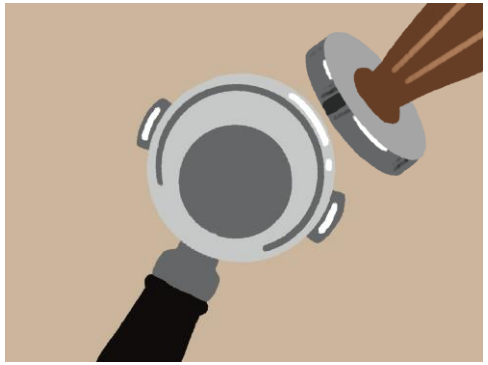
For the first time using Wekinator in this course, I found it quite challenging to work with. However, after completing several projects, I realized that Wekinator is an attractive method when making creative projects.

So for my final project, I really wanted to make something using Wekinator.

- **To make ‘game’**

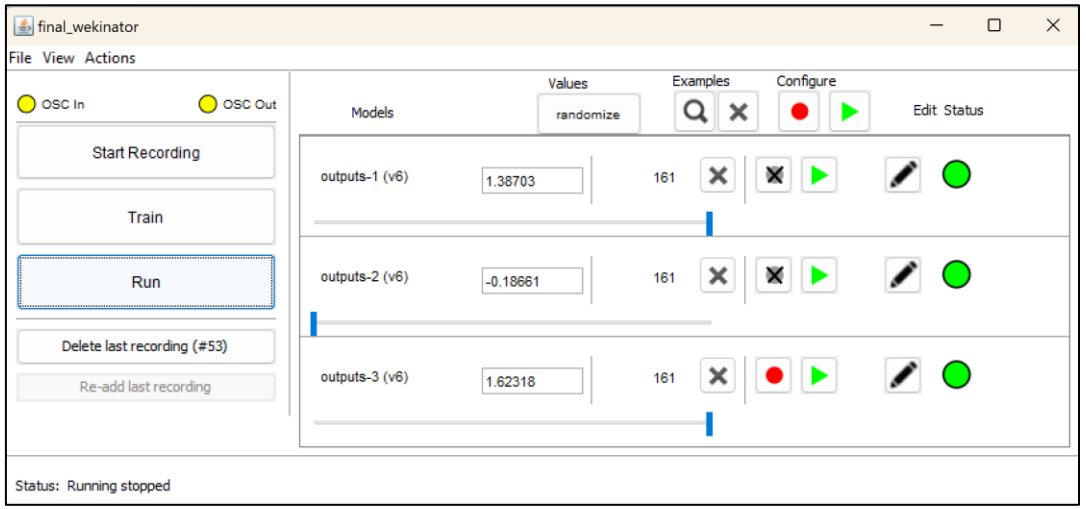
While I have created a few “games” in previous projects, I wanted to develop a truly fun and addictive game this time. And I suddenly remembered the flash game **고향만두**. It’s a game that is both fun and fairly challenging, making players want to keep trying. Inspired by this, I came up with the idea of a game that evaluates how well the player makes coffee, with different inputs at each stage.

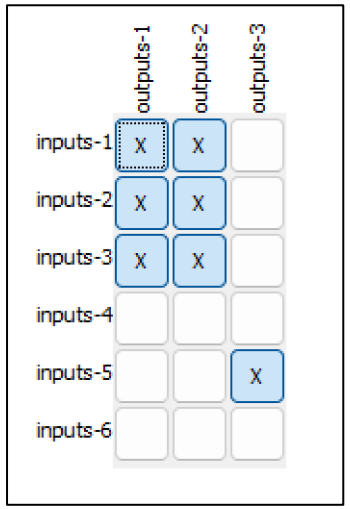
What I made



How I made

- **Wekinator**





Inputs : 6 (Acc, Att)
Outputs : 3

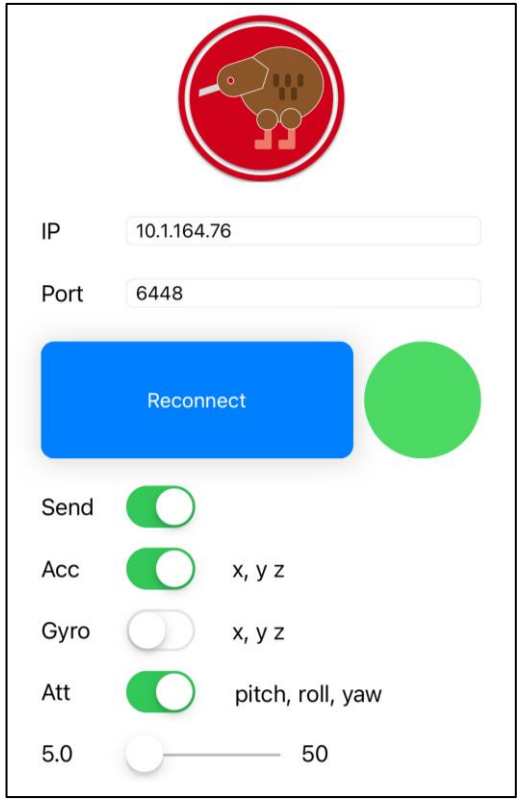
Outputs-1 (Roasting)
: Detects x, y, z Acc values

Outputs-2 (Grinding)
: Detects x, y, z Acc values

Outputs-3 (Pouring)
: Detects roll Att value

How I made

- **MotionSender**



Send Acc & Att

Slow down the detecting frame
to reduce noise

How I made

0. Setup()

```
oscP5 = new OscP5(this, 12000);
wekinator = new NetAddress("127.0.0.1", 6448);

// bgm = new SoundFile(this, "sounds/bgm.mp3");
bgm = new SoundFile(this, "sounds/copyright_free_bgm.mp3");
picking = new SoundFile(this, "sounds/picking.mp3");
roasting = new SoundFile(this, "sounds/roasting.mp3");
grinding = new SoundFile(this, "sounds/grinding.mp3");
tamping = new SoundFile(this, "sounds/tamping.mp3");
pouring = new SoundFile(this, "sounds/pouring.mp3");

bgm.loop();
bgmPlaying = true;

start =  loadImage("images/start.png");
tree = loadImage("images/tree.png");
bean1 = loadImage("images/bean1.png");
bean2 = loadImage("images/bean2.png");
bean3 = loadImage("images/bean3.png");
pan = loadImage("images/pan.png");
bean_unbaked = loadImage("images/bean_unbaked.png");
bean_baked1 = loadImage("images/bean_baked1.png");
bean_baked2 = loadImage("images/bean_baked2.png");
fire = loadImage("images/fire.png");
potterfilter = loadImage("images/potterfilter.png");
tamper1 = loadImage("images/tamper1.png");
tamper2 = loadImage("images/tamper2.png");
pressed1 = loadImage("images/pressed1.png");
pressed2 = loadImage("images/pressed2.png");
flowing = loadImage("images/flowing.png");
watercup = loadImage("images/watercup.png");
shot1 = loadImage("images/shot1.png");
shot2 = loadImage("images/shot2.png");
coffee = loadImage("images/coffee.png");
message = loadImage("images/message.png");
```

- OSC setting for wekinator
- Load sound files
- Load images

0. oscEvent()

```
void oscEvent(OscMessage msg) {
  if (msg.checkAddrPattern("/wek/outputs")) {
    output1 = msg.get(0).floatValue(); // roasting
    output2 = msg.get(1).floatValue(); // grinding
    output3 = msg.get(2).floatValue(); // pouring

    // println("Output1: " + output1 + ", Output2: " + output2 + ", Output3: " + o

    if (stage == 2 && output1 > 4.5) {
      roastLevel++;
      fired = true;
    }
    else fired = false;

    if (stage == 3 && output2 > 4.5) {
      grinding.play();
      grindLevel++;
      powders.clear();
      for (int i = 0; i < max_powder; i++) {
        powders.add(new Powder(random(280,480), random(196,396), random(1, 5)));
      }
    }
    else {
      grinding.stop();
    }

    if (stage == 6 && output3 > 4.5) {
      poured = true;
    }


  }
}
```

How I made

1. Function for start stage

```
void drawStart() {
  image(start, 0, 0, width, height);

  bgmPlaying = true;

  score = 100;
  roastLevel = 0;
  poured = false;
  grindLevel = 0;
  tamped = false;
  allBeansPicked = false;
  show_unbaked_bean = true;
  show_baked_bean = false;
  show_burned_bean = false;
  show_tamper1 = false;
  show_tamper2 = false;

  // start button
  textAlign(CENTER, CENTER);
  fill(190, 190, 160);
  strokeWeight(5);
  stroke(140, 140, 110);
  rect(550, 500, 180, 80);
  fill(140, 140, 110);
  text("Start", 640, 540);

  if (mousePressed && (mouseX > 550 && mouseX < 730 && mouseY > 500 && mouseY < 580)) {
    show1 = true;
    show2 = true;
    show3 = true;
    pickStart = millis();
    stage = 1;
  }
}
```

- Reset all variables
- If start button is clicked, progress to next stage

2. Function for picking beans stage

```
void drawPickBeans() {
  image(tree, 0, 0, width, height);

  fill(0);
  textAlign(LEFT, UP);
  text("1단계 : 커피콩 따기", 30, 50);
  text("마우스를 클릭해 콩을 따세요.", 30, 100);


  if (show1) image(bean1, 0, 0, width, height);
  if (show2) image(bean2, 0, 0, width, height);
  if (show3) image(bean3, 0, 0, width, height);

  if (mousePressed) {
    if (mouseX > 180 && mouseX < 255 && mouseY > 210 && mouseY < 300) {
      show1 = false;
      picking.play();
    }
    if (mouseX > 310 && mouseX < 373 && mouseY > 295 && mouseY < 363) {
      show2 = false;
      picking.play();
    }
    if (mouseX > 447 && mouseX < 510 && mouseY > 292 && mouseY < 346) {
      show3 = false;
      picking.play();
    }
  }

  if (millis() - pickStart >= 3000)  {
    if (show1 == false && show2 == false && show3 == false) allBeansPicked = true;
    else {
      allBeansPicked = false;
      score -= 10;
    }

    roastStart = millis();
    stage = 2;

  }
  //text(mouseX, mouseX, mouseY - 10);
  //text(mouseY, mouseX, mouseY + 10);
}
```

- Show beans' images
- Delete those images when clicked

How I made

3. Function for roasting stage

```
void drawRoastBeans() {

  image(pan, 0, 0, width, height);

  fill(0);
  textAlign(LEFT, UP);
  text("2단계: 커피콩 볶기", 30, 50);
  text("휴대폰을 위아래로 흔들어 커피콩을 볶으세요!", 30, 100);

  textAlign(CENTER, CENTER);

  text("로스팅 정도: " + roastLevel, width / 2, height - 50);

  if (fired) {
    roasting.play();
    image(fire, 0, 0, width, height);
  }
  else roasting.stop();

  if (show_unbaked_bean) image(bean_unbaked, 0, 0, width, height);
  if (show_baked_bean) image(bean_baked1, 0, 0, width, height);
  if (show_burned_bean) image(bean_baked2, 0, 0, width, height);


  if (roastLevel >= 10) {
    show_unbaked_bean = false;
    show_baked_bean = false;
    show_burned_bean = true;
  }
  else if (roastLevel >= 5) {
    show_unbaked_bean = false;
    show_baked_bean = true;
    show_burned_bean = false;
  }
```

```
  if (millis() - roastStart >= 5000) {
    roasting.stop();
    if (roastLevel < 5) score -= 20;
    else if (roastLevel >= 10) score -= 10;
    grindStart = millis();
    stage = 3;
  }

}
```

- Show different color of roasted bean, based on the roastLevel.

*oscEvent : detecting roasting, if true / roastLevel++;

4. Function for grinding stage

```
void drawGrindBeans() {
  image(potterfilter, 0, 0, width, height);

  fill(0);
  textAlign(LEFT, UP);
  text("3단계: 커피콩 갈기", 30, 50);
  text("휴대폰을 이용해 커피콩을 갈아주세요!", 30, 100);

  for (int i = 0; i < powders.size(); i++) {
    powders.get(i).render();
  }


  if (millis() - grindStart >= 3000) {
    grinding.stop();
    if (grindLevel < 5) score -= 10;
    tampStart = millis();
    show_tamper1 = true;
    stage = 4;
  }
}
```

*oscEvent : detecting grinding, if true / grindLevel++;

How I made

5. Function for tamping stage

```
void drawTamp() {
  image(potterfilter, 0, 0, width, height);
  if (show_tamper1) image(tamper1, 0, 0, width, height);

  if (roastLevel >= 10) image(pressed2, 0, 0, width, height);
  else image(pressed1, 0, 0, width, height);

  if (show_tamper2) image(tamper2, 0, 0, width, height);

  fill(0);
  textAlign(LEFT, UP);
  text("4단계: 탬핑", 30, 50);
  text("마우스를 클릭해 탬핑하세요!", 30, 100);


  if (mousePressed) {
    tamping.play();
    tamped = true;
    show_tamper1 = false;
    show_tamper2 = true;
  }

  if (millis() - tampStart >= 3000) {
    tamping.stop();
    if (tamped == false) score -= 20;
    flowingStart = millis();
    stage = 5;
  }
}
```

6. Function for extracting stage

```
void flowing() {
  image(flowing, 0, 0, width, height);

  fill(0);
  textAlign(LEFT, CENTER);
  text("추출 중...", 600, 300);

  if (millis() - flowingStart >= 3000) {
    pourStart = millis();
    stage = 6;
  }
}
```

7. Function for pouring stage

```
void drawPourCoffee() {
  image(watercup, 0, 0, width, height);
  if (show_shot1) image(shot1, 0, 0, width, height);
  if (show_shot2) image(shot2, 0, 0, width, height);

  fill(0);
  textAlign(LEFT, UP);
  text("5단계: 커피 따르기", 30, 50);
  text("휴대폰을 기울여 커피를 따라 주세요!", 30, 100);

  if (poured) {
    pouring.play();
    show_shot2 = true;
    show_shot1 = false;
  }
  else {
    show_shot1 = true;
    show_shot2 = false;
  }


  if (millis() - pourStart >= 2000) {
    pouring.stop();
    if (!poured) score -= 50;
    stage = 7;
  }
}
```

oscEvent : detecting pouring,
if true / poured = true;

How I made

8. Function for result

```
void drawResult() {
  if (poured) image(coffee, 0, 0, width, height);
  else image(watercup, 0, 0, width, height);

  image(message, 0, 0, width, height);

  fill(0);
  textAlign(LEFT, UP);
  text(":: 결과 :: ", 30, 50);

  textAlign(CENTER, CENTER);
  float cx = 600;
  float cy = 200;


  if (!poured) {
    text("엑!! 맹물이잖아!", cx, cy - 20);
  } else if (roastLevel < 5 || roastLevel > 15) {
    text("맛없어...", cx, cy - 20);
  } else if (!allBeansPicked) {
    text("밍밍해...", cx, cy - 20);
  } else if (grindLevel < 5) {
    text("커피가 덜 갈렸어...", cx, cy - 20);
  } else if (!tamped) {
    text("흠... 탬핑이 부족해", cx, cy - 20);
  } else {
    text("오...", cx, cy - 20);
  }
  text("점수는..." + score + "점!!!", cx, cy + 20);
```

- Show short evaluation with score
- Show restart button

```
  // restart button
  textAlign(CENTER, CENTER);
  fill(190, 190, 160);
  strokeWeight(5);
  stroke(140, 140, 110);
  rect(500, 440, 180, 80);
  fill(140, 140, 110);
  text("Restart", 590, 480);

  if (mousePressed && (mouseX > 500 && mouseX < 680 && mouseY > 440 && mouseY < 520)) {
    bgmPlaying = false;
    stage = 0;
  }
}
```

Links

- zip file for my project

https://drive.google.com/file/d/1PQAOA_Xf6z9xpcUIvkSuU0eJtEQex3Wb/view?usp=sharing

- video

https://drive.google.com/file/d/1__Yp2Xv0f9nTW7I-0RAFLH4IHA8wOGLm/view?usp=sharing